SmartRoute AI: An Explainable Risk-Aware Deep Reinforcement Learning Framework for Dynamic Last-Mile Delivery Assignment

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Abstract

Last-mile delivery is a critical stage of logistics where assigning delivery orders to suitable delivery partners efficiently is challenging due to changing traffic conditions, delivery deadlines, partner availability, distance, and operational risks. This project proposes **SmartRoute AI**, an explainable and risk-aware deep reinforcement learning framework for dynamic last-mile delivery assignment. The system continuously analyzes delivery requests, delivery-partner availability, traffic conditions, distance, estimated travel time, and delivery priorities. A Deep Reinforcement Learning (DRL) agent learns to select suitable delivery partners for incoming orders based on the current logistics environment. The framework incorporates risk factors such as traffic congestion, delayed deliveries, excessive travel distance, and partner workload into the decision-making process. Unlike conventional assignment methods, the proposed system dynamically adapts to changes in real-time delivery conditions. The Hungarian Algorithm can be integrated as a baseline or optimization component for comparing assignment performance. An explainable AI layer provides understandable reasons for each assignment decision, such as lower travel time, reduced risk, or better workload balance. The system aims to minimize delivery delays, travel distance, and operational risk while improving partner utilization. A simulation-based environment can be used to train and evaluate the DRL model without requiring physical hardware. Performance can be measured using delivery time, assignment accuracy, total distance, workload distribution, and risk reduction. The proposed framework provides a scalable software-based solution for intelligent, adaptive, and explainable last-mile delivery management.

Keywords— Deep Reinforcement Learning, Last-Mile Delivery, Dynamic Assignment, Explainable AI, Risk-Aware Optimization.